APPENDIX A: Theoretical framework

In Appendix a, we have presented all the calculations performed, equation by equation, in the development of the model in Chapter 3 (theoretical framework).

A3.2 Free Market Scenario

Equation 1

$$Y_r = A_r K_r^{(1-\alpha)} E_r^{\alpha}$$

Equation 2

Let's maximize the profit function:

$$\max_{w_r, r_r} A_r K_r^{(1-\alpha)} E_r^{\alpha} - w_r L_r - r_r K_r$$

First, we differentiate with respect to the wage, then with respect to the return to capital, and we obtain the demand for labour and the demand for capital:

$$w_r = \alpha A_r K_r^{(1-\alpha)} E_r^{(\alpha-1)}$$

$$r_r = (1-\alpha) A_r K_r^{-\alpha} E_r^{\alpha}$$

Equation 3

We apply the logarithm to both:

$$\ln w_r = \ln \alpha + \ln A_r + (1-\alpha) \ln K_r - (1-\alpha) \ln E_r$$

$$i_r \equiv \ln r_r = \ln(1-\alpha) + \ln A_r - \alpha \ln K_r + \alpha \ln E_r$$

Now, we know that the supply of capital is respectively:

$$ir = \mu \ln K_r$$

Write the equilibrium and apply the logarithm:

$$\frac{w_n^*}{p_n^\sigma} = \frac{w_s^*}{p_s^\sigma}$$

$$\ln w_n - \sigma \ln p_n = \ln w_s - \sigma \ln p_s \qquad \ln w_n - \sigma\gamma \ln E_n = \ln w_s - \sigma\gamma \ln E_s$$

In equilibrium supply is equal to demand (of capital, in this case):

$$\mu \ln K_r = \ln(1-\alpha) + \ln A_r - \alpha \ln K_r + \alpha \ln E_r$$

Noe let's isolate $\ln K_r$:

$$\ln K_r = \frac{1}{\mu+\alpha}[\ln(1-\alpha) + \ln A_r + \alpha \ln E_r]$$

Plug $\ln K_r$ in $\ln w_r$, we obtain:

$$\ln w_r = \ln \alpha + \ln A_r + \frac{1-\alpha}{\mu+\alpha}[\ln(1-\alpha) + \ln A_r + \alpha \ln E_r] - (1-\alpha)\ln E_r$$

Now we plug this equation in the equilibrium condition:

$$\ln w_n - \sigma \ln p_n = \ln w_s - \sigma \ln p_s$$

$$\ln \alpha + \ln A_n + \frac{1-\alpha}{\mu+\alpha}[\ln(1-\alpha) + \ln A_n + \alpha \ln E_n] - (1-\alpha)\ln E_n - \sigma \ln p_n =$$
$$= \ln \alpha + \ln A_s + \frac{1-\alpha}{\mu+\alpha}[\ln(1-\alpha) + \ln A_s + \alpha \ln E_s] - (1-\alpha)\ln E_s - \sigma \ln p_s$$

Remember that $\ln p_r = \gamma \ln L_r = \gamma \ln E_r$:

$$\ln \alpha + \ln A_n + \frac{1-\alpha}{\mu+\alpha}[\ln(1-\alpha) + \ln A_n + \alpha \ln E_n] - (1-\alpha)\ln E_n - \sigma\gamma \ln E_n$$
$$= \ln \alpha + \ln A_s + \frac{1-\alpha}{\mu+\alpha}[\ln(1-\alpha) + \ln A_s + \alpha \ln E_s] - (1-\alpha)\ln E_s - \sigma\gamma \ln E_s$$

Finally isolate $\ln E_n - \ln E_s$:

$$\ln E_n - \ln E_s = \frac{1+\mu}{\sigma\gamma(\mu+\alpha)+\mu(1-\alpha)}\,(\ln A_n - \ln A_s) > 0$$

<u>*Equation 4*</u>

Equalize supply and demand of capital:

$$\ln K_r = \ln(1-\alpha) + \ln A_r - \alpha \ln K_r + \alpha \ln E_r$$

Isolate $\ln K_n - \ln K_s$:

$$\ln K_n - \ln K_s = \frac{1}{\mu+\alpha}[(\ln A_n - \ln A_s) + \alpha(\ln E_n - \ln E_s)]$$

Plug equation 3 in the previous one:

$$\ln K_n - \ln K_s = \frac{1}{\mu+\alpha}[(\ln A_n - \ln A_s) + \frac{\alpha(1+\mu)}{\sigma\gamma(\mu+\alpha)+\mu(1-\alpha)}\,(\ln A_n - \ln A_s)]$$

After some calculation:

$$\ln K_n - \ln K_s = \frac{\sigma\gamma+1}{\sigma\gamma(\mu+\alpha)+\mu(1-\alpha)}\,(\ln A_n - \ln A_s) > 0$$

<u>*Equation 5*</u>

We take the equilibrium function and isolate $\ln w_n - \ln w_s$:

$$\ln w_n - \sigma\gamma \ln E_r = \ln w_s - \sigma\gamma \ln E_r$$
$$\ln w_n - \ln w_s = \sigma\gamma(\ln E_r - \ln E_r)$$

Plug equation 3 in the previous one:

$$\ln w_n - \ln w_s = \frac{\sigma\gamma(1+\mu)}{\sigma\gamma(\mu+\alpha)+\mu(1-\alpha)}\,(\ln A_n - \ln A_s) > 0$$

Equation 6

We consider the equilibrium function, and we isolate $\ln p_n - \ln p_s$:

$$\ln w_n - \sigma \ln p_n = \ln w_s - \sigma \ln p_s$$
$$\ln p_n - \ln p_s = \frac{1}{\sigma}(\ln w_n - \ln w_s)$$

Plug equation 5 in the previous one:

$$\ln p_n - \ln p_s = \frac{\gamma(1+\mu)}{\sigma\gamma(\mu+\alpha)+\mu(1-\alpha)}(\ln A_n - \ln A_s) > 0$$

A3.3 Fixed-wage case

Equation 7

Now salaries are exogenous:

$$\overline{w} = w_n^* > w_s^*$$

Equation 8

Similarly to the free market case, we start with the production function and find the demand for labor (i.e., the wage) and the demand for capital. Then, we isolate $\ln w_r$:

$$Y_r = A_r K_r^{(1-\alpha)} E_r^{\alpha} \qquad w_r = \alpha A_r K_r^{(1-\alpha)} E_r^{(\alpha-1)} \qquad r_r = (1-\alpha) A_r K_r^{-\alpha} E_r^{\alpha}$$

$$\ln w_r = \ln \alpha + \ln A_r + (1-\alpha)\ln K_r - (1-\alpha)\ln E_r$$

Then, we equate the demand and supply of capital and obtain:

$$\ln K_r = \frac{1}{\mu+\alpha}[\ln(1-\alpha) + \ln A_r + \alpha \ln E_r]$$

Plugging $\ln K_r$ in $\ln w_r$. Now, we isolate $\ln w_n - \ln w_s$:

$$\ln w_n - \ln w_s = \left[1 + \frac{1-\alpha}{\mu+\alpha}\right] (\ln A_n - \ln A_s) + \left[\frac{\alpha(1-\alpha)}{\mu+\alpha} - (1-\alpha)\right] (\ln E_n - \ln E_s)$$

Since $\bar{w} = w_n = w_s$, $\ln w_n - \ln w_s = 0$:

$$\left[1 + \frac{1-\alpha}{\mu+\alpha}\right] (\ln A_n - \ln A_s) + \left[\frac{\alpha(1-\alpha)}{\mu+\alpha} - (1-\alpha)\right] (\ln E_n - \ln E_s) = 0$$

Finally, we isolate $\ln E_n - \ln E_s$:

$$\ln E_n - \ln E_s = \frac{1+\mu}{(1-\alpha)\mu} (\ln A_n - \ln A_s) > 0$$

<u>Equation 8a</u>

We know that $w_n = w_s$, so:

$$\alpha\, A_n\, K_n^{(1-\alpha)} E_n^{(\alpha-1)} = \alpha\, A_s\, K_s^{(1-\alpha)} E_s^{(\alpha-1)}$$

Then we isolate $\frac{K_n}{K_s}$ and then we apply the logarithm:

$$\ln K_n - \ln K_s = -\frac{1}{1-\alpha} (\ln A_n - \ln A_s) + (\ln E_n - \ln E_s)$$

Plug equation 8 into the previous one:

$$\ln K_n - \ln K_s = -\frac{1}{1-\alpha} (\ln A_n - \ln A_s) + \frac{1+\mu}{(1-\alpha)\mu} (\ln A_n - \ln A_s)$$

Just few calculation:

$$\ln K_n - \ln K_s = \frac{1}{(1-\alpha)\mu} (\ln A_n - \ln A_s) > 0$$

Equation 9

We start from the equilibrium condition:

$$\Omega_n = \Omega_s$$

$$\frac{\overline{w}(1-u_n^{**})^{\delta}}{p_n^{**\sigma}} = \frac{\overline{w}(1-u_s^{**})^{\delta}}{p_s^{**\sigma}}$$

Since $\overline{w} = w_n^*$, $E_n^{**} = L_n^{**} \rightarrow u_n^{**} = 0$. We also divide by $\overline{w}$:

$$\frac{1}{p_n^{**\sigma}} = \frac{(1-u_s^{**})^{\delta}}{p_s^{**\sigma}}$$

Isolating u_s^{**}we obtain:

$$u_s^{**} = 1 - \left(\frac{L_s}{L_n}\right)^{\frac{\gamma\sigma}{\delta}}$$

Now we should find $\frac{L_s}{L_n}$ in function of A_s and A_n. Before we find $\frac{L_s}{L_n}$ in function of E_s and E_n and then we exploit the equation 8:

$$\Omega_n = \Omega_s$$

$$\frac{1}{p_n^{**\sigma}} = \frac{(1-u_s^{**})^{\delta}}{p_s^{**\sigma}}$$

$$\frac{1}{L_n^{**\sigma\gamma}}\left(\frac{E_n}{L_n}\right)^{\delta} = \frac{1}{L_s^{**\sigma\gamma}}\left(\frac{E_s}{L_s}\right)^{\delta}$$

$$\frac{L_s^{**}}{L_n^{**}} = \left(\frac{E_n^{**}}{E_s^{**}}\right)^{-\frac{\delta}{\sigma\gamma+\delta}}$$

We plug this formula in $u_s^{**} = 1 - \left(\frac{L_s}{L_n}\right)^{\frac{\gamma\sigma}{\delta}}$:

$$u_s^{**} = 1 - \left(\frac{E_n^{**}}{E_s^{**}}\right)^{-\frac{\gamma\sigma}{\gamma\sigma+\delta}}$$

We apply the logarithm:

$$\ln u_s^{**} = \frac{\gamma\sigma}{\gamma\sigma + \delta}(\ln E_n^{**} - \ln E_s^{**})$$

Plug equation 8 in the last one:

$$\ln u_s^{**} = \frac{(1+\mu)\sigma\gamma}{\mu(1-\alpha)(\sigma\gamma+\delta)}\ (\ln A_n - \ln A_s) > 0$$

Since $\ln(1+x) \sim x$ for small x, we can finally write:

$$u_s^{**} = \frac{(1+\mu)\sigma\gamma}{\mu(1-\alpha)(\sigma\gamma+\delta)}\ (\ln A_n - \ln A_s) > 0$$

Equation 10

We start from the equilibrium and isolate $1 - u_s^{**}$:

$$\frac{\overline{w}}{p_n^{**\sigma}} = \frac{\overline{w}(1-u_s^{**})^{\delta}}{p_s^{**\sigma}}$$

$$\left[\frac{\overline{w}}{p_n^{**\sigma}}\left(\frac{\overline{w}}{p_s^{**\sigma}}\right)^{-1}\right]^{\frac{1}{\delta}} = 1 - u_s^{**}$$

Applying the logarithm we have:

$$\frac{1}{\delta}\left(\ln\frac{\overline{w}}{p_n^{**\sigma}} - \ln\frac{\overline{w}}{p_s^{**\sigma}}\right) = \ln 1 - \ln u_s^{**}$$

Plug equation 9 in the previous one and we get:

$$\ln\frac{\overline{w}}{p_n^{**\sigma}} - \ln\frac{\overline{w}}{p_s^{**\sigma}} = -\frac{(1+\mu)\sigma\delta\gamma}{\mu(1-\alpha)(\sigma\gamma+\delta)}\ (\ln A_n - \ln A_s) < 0$$

Since $\ln(1+x) \sim x$ for small x, we can finally write:

$$\frac{\overline{w}}{p_n^{**\sigma}} - \frac{\overline{w}}{p_s^{**\sigma}} = -\frac{(1+\mu)\sigma\delta\gamma}{\mu(1-\alpha)(\sigma\gamma+\delta)}\ (\ln A_n - \ln A_s) < 0$$

Equation 10a

We consider the equilibrium, and we isolate $\ln p_n^{**} - \ln p_s^{**}$:

$$\Omega_n = \Omega_s \qquad \frac{1}{p_n^{**\sigma}} = \frac{(1-u_s^{**})^{\delta}}{p_s^{**\sigma}} \rightarrow \ln p_n^{**} - \ln p_s^{**} = \frac{\delta}{\sigma} \ln u_s^{**}$$

We plug equation 9 in the previous one:

$$\ln p_n^{**} - \ln p_s^{**} = \frac{(1+\mu)\delta\gamma}{\mu(1-\alpha)(\sigma\gamma+\delta)} (\ln A_n - \ln A_s) > 0$$

A3.4 Aggregate Effects

Equation 11

In equilibrium, demand is equal to supply (of capital):

$$ir = \mu \ln K_r$$
$$\mu \ln K_r = \ln(1-\alpha) + \ln A_r - \alpha \ln K_r + \alpha \ln E_r$$

Isolate $\ln K_r$ and change index with s:

$$\ln K_s = \frac{1}{\mu+\alpha} [\ln(1-\alpha) + \ln A_s + \alpha \ln E_s]$$

Equation 12

We start from this assumption $Y_n^* + Y_s^* > Y_n^{**} + Y_s^{**}$:

$$A_n^* K_n^{*(1-\alpha)} E_n^{*\alpha} + A_s^* K_s^{*(1-\alpha)} E_s^{*\alpha} = A_n^{**} K_n^{**(1-\alpha)} E_n^{**\alpha} + A_s^{**} K_s^{**(1-\alpha)} E_s^{**\alpha}$$

Now we consider the equation of the salary $w_r = \alpha A_r K_r^{(1-\alpha)} E_r^{(\alpha-1)}$ and we notice that:

$$\frac{w_r}{\alpha E_r^{(\alpha-1)}} = A_r K_r^{(1-\alpha)}$$

Now we plug in this into the previous equation paying attention to the index and the asterix:

$$\frac{w_n^*}{\alpha E_n^{*\,(\alpha-1)}} E_n^{*\alpha} + \frac{w_s^*}{\alpha E_s^{*\,(\alpha-1)}} E_s^{*\alpha} > \frac{w_n^{**}}{\alpha E_n^{**\,(\alpha-1)}} E_n^{**\alpha} + \frac{w_s^{**}}{\alpha E_s^{**\,(\alpha-1)}} E_s^{**\alpha}$$

$$\frac{1}{\alpha}[w_n^* E_n^* + w_s^* E_s^*] > \frac{1}{\alpha}[w_n^{**} E_n^{**} + w_s^{**} E_s^{**}]$$

Finally, I divide by $\bar{L}$ and multiply by α (notice that $w_n^{**} = w_s^{**} = \bar{w}$):

$$\frac{[w_n^* E_n^* + w_s^* E_s^*]}{\bar{L}} > \frac{[\bar{w} E_n^{**} + \bar{w} E_s^{**}]}{\bar{L}}$$

Equation 13

Since we know that $w_n^* = \bar{w}$ and $E_n^* = E_n^{**}$, we can rewrite the equation 11 in this way.

$$\frac{w_s^* E_s^*}{\bar{L}} > \frac{\bar{w} E_s^{**}}{\bar{L}}$$

Since $E_s^{**} = L_s^{**}(1 - u_s^{**})$ and $L_s^{**} = E_s^*$:

$$\frac{w_s^* E_s^*}{\bar{L}} > \frac{\bar{w} E_s^*}{\bar{L}}(1 - u_s^{**})$$

We proceed with the calculations:

$$\frac{w_s^* E_s^*}{\bar{L}} > \frac{\bar{w} E_s^*}{\bar{L}} - \frac{\bar{w} E_s^*}{\bar{L}} u_s^{**}$$

$$E_s^*(\bar{w} - w_s^*) < \frac{\bar{L}\bar{w} E_s^* u_s^{**}}{\bar{L}}$$

Since $\bar{L}$ in the denominator is referred to the population in s, we can simplify with E_s^*. We can do it because in the free-market case there is no unemployment:

$$(\bar{L} u_s^{**})\bar{w} > E_s^*(\bar{w} - w_s^*)$$

Appendix B: Data Description

B.1 Dataset description

We have used 3 files for our analysis: “provinces.dta” (for Chapter 4 and 5), “bargaining.dta” (for chapter 2) and “amenities.dta” (for chapter 6).

The file “province_data.dta” contains the main variables like wages adjusted for workforce composition, CPI and house price indices, real wages, non-employment rate, rate of irregular employment and gross value added per worker. It is a big dataset that contain 22 variables and 4255 observations both for Italy and Germany.

The file “bargaining.dta” is aggregated data on industry contracts and opening clauses based on the IAB Establishment Panel from 1996 to 2013. This study uses the anonymous sample of integrated labour market biographies from 1975 to 2014. This dataset contains 369 observations and 66 variables on years, German regions, industrial contract and opening clauses under diverse tariff regime (un\corrected and no\weighted). It contributes to analyse the German wage setting mechanism.

The file “amenities.dta” contains data on indicators of amenities by Italian province. These data are based on the yearly Sole24ore “Indagine sulla qualità della vita”. This dataset has 1030 observations and 10 variables on climate, air pollution, population density, crimes, school and health system. Regressions on this dataset yield important results on the impact of local services on Italian internal migration. Table 11 is taken from ISTAT 2019 Analysis of the Determinants of Life Satisfaction.

B.2 Geographic Unit of Analysis

It is crucial to correctly define the geographical units of local labour markets. In our analysis, the geographical units are the provinces, because they have an appropriate size to be identified as individual open economies. Regions are too large entities to capture local variety and complexity, while, on the contrary, municipalities are too small, as the labour force very often works in municipalities neighbouring their own residence.

Italy is divided into 103 regions, 62 of which are considered in our analysis from the North and 41 from the South. The 62 northern provinces are in the following regions: Emilia-Romagna, Friuli-Venezia Giulia, Liguria, Lombardia, Marche, Piemonte, Toscana, Trentino-Alto Adige, Umbria, Valle d'Aosta and Veneto. The 41 Southern provinces are in the following regions: Lazio, Campania, Molise, Basilicata, Puglia,

Calabria, Sicilia and Sardegna. Of the 39,028,260 Italians of working age, 21,507,750 reside in the North and the other 17,520,510 reside in the South. The average is 495,104 people per province (in 2010).

Germany is divided into 96 province ('Raumordnungsregions'), 74 of which are in the West and 22 in the East. The 'Lander' that were in the GDR before the fall of the Berlin Wall are considered East, those that were in West Germany are considered West. The average working age is 737,448 persons per province (in 2010).

B.3 Productivity (gross value added)

The only available detailed measure for firms' productivity is the gross value added. It is the value is of firms' production minus estimated inputs at selling prices. the basis of the price is a unit of a product or service, minus rates and subsidies on the product. One way to define its formula is:

$$GVA = GDP + SP - TP$$

where GDP is the gross domestic product, SP are subsidies on product and TP are taxes on products.

The average value added per worker is slightly higher in Italy. But the hourly value added is higher in Germany. The reason is that, since the amount of part time workers is bigger in Germany (26.3%) then Italy (15%), the average value added per worker is slightly higher in Italy. The highest shares of part-time work belong to the most productive regions, but the dispersion of the fraction of temporary workers across local areas is similar between the two countries. This implies that the relationship between value added and temporary employment between the provinces is very similar in the two countries and this is not a problem for our analysis.

B.4 Nominal wages

For Italy, wages data come from quarterly statistics of ISTAT (Istituto Nazionale di Statistica). Information is available only from 2009 to 2013. Respondents give information about net monthly wage and average hours worked in the previous four weeks. That allows us to figure out the conditional mean wages. For Germany, wages

data come from quarterly statistics of IAB for Germany (Institut für Arbeitsmarkt). We have more data available (from 1992 to 2014), but we consider the same time window 2009-2013. Individual-level daily wages are obtained by SIAB. In both countries, only full-time wages are considered, excluding all non-recurring components of pay.

After this preamble, we measure the mean wages for province and year in Italy and in Germany, pay attention to the worker specialization and sectorial effects between provinces. To estimate the conditional mean wages net of worker characteristics and sectoral effects, we use individual-level data for Italy and Germany to regress:

$$w_i = \alpha + Z_i\beta + u_i$$

where w_i is the hourly wage (Italy) or daily wage (Germany) of worker i and Z_i is a vector that includes gender, age, age squared, education, and sector of worker i. The regression is run separately for Italy and Germany.

B.5 Employment and Informal Sector

German employment rate is calculated IAB and in Italy by ISTAT. These institute are the fundamental source on which this dataset is built. Employment rates refer to workers from 15 to 64 years old. Differently from data derived from social security records, Italian employment rate comes from anonymous survey. This because there is a huge informal sector in Italy. Therefore, workers in the sample have limited incentives to misreport their employment status. However, this measure is not sufficient to solve the problem of informal sector in Italy. South of Italy suffers more from this problem. Thus, we risk overestimating southern non-employment rate. To overcome this problem we will use employment data already corrected by informal work. It is impressive the difference between the Italian corrected 34% and uncorrected 42.5% non-employment rate. The unemployment rate in Germany is 27.9%.

B.6 Housing Cost

All the data concerning Italian transaction on residential property sales were taken from “Osservatorio del Mercato Immobiliare Italiano” between 2004 and 2011. We regress:

$$p_i = \alpha + X_i\beta + u_i$$

where p_i is the price per square meter of housing unit i and X_i is a vector that includes the size of the unit, presence of a balcony, terrace, or cellar, brightness, view, orientation, measures of construction quality, distance to retail stores, distance to public transportation, distance to public parks or gardens, parking, indicators for the type of neighborhood, and whether the municipality is near touristic area.

For what concern Germany, data on housing price are taken from BBSR (Bundesinstitut für Bau Stadt und Raumforschung). They consist in a data collection on online or newspaper ads about rent prices between 2004 and 2014. To calculate prices per square meter, BBSR uses unfurnished apartments of a size between 40 and 130 square meters listed for less than half a year. We compute weighted averages for Raumordnungsregion, using weights reflecting the housing stock in the area.

In the next subchapter, we look for the local cost of living index. However, in the theoretical analysis we considered only the housing price.

B.7 Local Cost of Living Index (CPI)

The statistical offices of both countries provide regional CPI measures that can be used to quantify the relationship between house prices and the cost of life. CPIs measure differences in the cost of living between provinces within each of the two countries. The local CPI_{pt} is defined as:

$$CPI_{pt} = \omega HP_{pt} + (1 - \omega)NHP_{pt}$$

where HP_{pt} is housing price in province p and year t, NHP_{pt} is the price of non-housing or non-tradable, and (ω) is the housing weight. Some part of NHP varies with the housing price so that: $NHP = \pi HP + v$. Therefore, when we regress ΔCPI_{pt} on ΔHP_{pt}: $\beta = (\omega + (1 - \omega)\pi)$. Thus, CPI_{pt} is directly proportional on housing price and NHP_{pt} that, in turn, is directly proportional to housing price again. Therefore, the qualitative outcomes of the theoretical framework would not change. So, our model results cleaner and simpler, and, at the same time, the empirical analysis will be more precise and in-depth. These data are important to find the real wages, and, we will see, as expected by the model, that real wages are higher in southern Italy (Moretti 2013).

Appendix C: Discrepant results

This part of the appendix describes how I recalculated all the calculations made in Boeri's paper. The purpose of recalculating everything related to the counterfactual calculations and empirical evidence is to understand in depth the quantitative significance of the paper. Stata and Excel were used to reproduce the calculations (within the limits of my skills).

The Excel file is intended to simplify the viewing and understanding of the calculations of the counterfactual exercise performed with Stata on the dataset 'province_data.dta' in the paper by Boeri, Moretti, Posch and Ichino (2021).

The results I obtained are slightly different in the values. Despite this, the concepts and conclusions remain the same. This small discrepancy (we are talking in the order of 2-3 percentage points) does not constitute any problem in the analysis and implications made in the paper.

C.1 Empirical Evidence

This part explains the calculation to retriever Table 3, which presents a summary of the most important empirical data from the analysis. These calculations are presented in the fourth and fifth sheets of the excel file. The fourth sheet presents data for Germany: from column A to J we have the variables country, high_gva, name_reg, name_prov, year, gva, wage_unif, empl_rate, cpi_moretti for East Germany; from column L to T, we have the same variables for West Germany.

At the end of columns G, H, I, J are the arithmetic averages of value added, nominal (daily) wage, employment rate and price level for all East German provinces. The same applies for all West provinces, considering columns Q, R, S, T. Finally, column V contains the values of the German averages weighted by the number of provinces for each macro-region of the value added, nominal (daily) wage, employment rate and CPI for the whole of Germany.

In the fifth sheet, the same is done for Italy, but only for the CPI, because the other values had already been calculated for the counterfactual analysis for table 7, that we deal with in the next subchapter.

C.2 Counterfactual Analysis

In the paper by Boeri et al (2021, p.38), it is explained that value added, wages and the employment rate of the median province in the north are used (also 2010 data). Confusion arises here, because we are not talking about the median values for the north, but about the values of the median provinces in the north. So at this point I followed two methodologies, which led me to slightly different results. Regarding the values for the median provinces in the north, I found these values by entering a code on stata:

tab gva if high_gva==1 | tab wage_orig if high_gva==1 | tab empl_orig if high_gva==1

After typing in these codes, an output table appears where we go to retrieve the values at the 50th percentile, i.e. the median. These values belong to the provinces of Pisa, Livorno and Padua, for value added, wage and employment rate respectively, which we have already presented in the sub-chapters 5.1, 5.2 e 5.3.

As far as the second methodology is concerned, I simply took Stata's output from the code into account:

Figure C2.1 Median of value added per worker, nominal wage and corrected employment rate of northern provinces.

Value added per worker

	Percentiles	Smallest		
1%	50929	50929		
5%	52894	51677		
10%	53813	51913	Obs	62
25%	55465	52894	Sum of wgt.	62
50%	59053.5		Mean	58576.29
		Largest	Std. dev.	4041.968
75%	61450	63687		
90%	63126	64510	Variance	1.63e+07
95%	63687	64581	Skewness	.7053009
99%	74226.32	74226.32	Kurtosis	4.963883

Nominal wage

	Percentiles	Smallest		
1%	8.064933	8.064933		
5%	8.156693	8.10336		
10%	8.311706	8.130368	Obs	62
25%	8.515545	8.156693	Sum of wgt.	62
50%	8.653431		Mean	8.663613
		Largest	Std. dev.	.283392
75%	8.81036	9.034279		
90%	8.911926	9.204414	Variance	.080311
95%	9.034279	9.278921	Skewness	.6580356
99%	9.753784	9.753784	Kurtosis	5.635141

Employment rate corrected

	Percentiles	Smallest		
1%	64.82192	64.82192		
5%	67.6177	65.07738		
10%	68.30427	65.92692	Obs	62
25%	69.49067	67.6177	Sum of wgt.	62
50%	71.23602		Mean	70.99252
		Largest	Std. dev.	2.493123
75%	72.39413	75.16989		
90%	73.8815	76.09176	Variance	6.215662
95%	75.16989	76.36607	Skewness	-.011486
99%	76.77625	76.77625	Kurtosis	3.248329

Trying both approaches, it emerged that it is more appropriate to use the median values for the northern provinces, rather than the median provinces (Pisa, Livorno and Padua).

In the first Excel sheet we calculate the values in the first panel, which concerns the average hourly wage. Since Equation 14 and 16 are used to calculate counterfactual wages, we need the elasticity of the wage with respect to value added, the value added of the median province of the north, all value added of the south, and the nominal wage of the median province of the north. All these values can be found in the output box on Stata when running the counterfactual calculation code. For the calculation we moved all this data to excel, entering wages in Status Quo in Column ‘M’, and then applied the following formula to find the Counterfactual I wages (in Column ‘N’):

=(1-(D4 * ((B4 - J2)/B4))) * C4

where, D4 = 0.739, B4 = 59053.5 and C4 = 8.653. This formula is then applied for the value added of all the southern provinces (J2 to J42, i.e. Lecce to Avellino). To arrive at final result, we calculated the weighted average of wages (in all three scenarios) per working age population in each province[1].

Same procedure for column 'O' of counterfactual II, except for the exclusion of the term 'D4' (i.e. elasticity) from the formula, since elasticity is equal to 1.

We thus obtain the results of the average hourly wage in the status quo and in the two counterfactuals, which we will also use to calculate the differences and all the values for Italy in general. Each result we found is the result of a weighted average for the population (working age), also to find the values for Italy, so we must also introduce the population of the northern provinces into the calculation file.

The columns 'Q' to 'V' take from the data editor the observations for each province in the North for the variables 'North' (this is the union of the variable country and the North indicator for Italy, country == 1 with high_gva == 1), 'name_reg' (the name of the region), 'cod_prov', 'name_prov' (the code and name of the provinces) and 'work_age_pop' (the working age population for each province in the North). The value-added variable ('gva_ncorr') is not needed here. We need the population variable to calculate weighted averages of Italian wages for the working-age population in the three scenarios[2]. Once the wages for Italy have been calculated, we can find the differences in euros and percentages. In the results of the first panel, we have a very small deviation

[1] $\overline{w}_s = \frac{\sum_{i=1}^{n} p_i w_i}{\sum_{i=1}^{n} w_i}$, where $\overline{w}_s$ is the southern weighted wage, w_i are the wage values for each province in South i, p_i are the weights associated with each value w_i, i.e. the working-age population of each province and, finally, n is the total number of provinces n = 41). As for the Status quo, this formula on Excel takes this form: "=SUMPRODUCT(K2:K42,M2:M42)/SUM(K2:K42)" (for English excel version; in Italian version is "=MATR.SOMMA.PRODOTTO(K2:K42; M2:M42)/SOMMA(K2:K42)"). For counterfactuals, we change "M" with "N", "O".

[2] All values are calculated with $\overline{w}_{ita} = \frac{\sum_{i=1}^{n} p_i w_i + \sum_{j=1}^{m} p_j w_j}{\sum_{i=1}^{n} w_i + \sum_{j=1}^{m} w_j}$ where w_i, p_i and n re the same value of the previous footnote, and w_j, p_j and m are the same value, bur for North. In other words, at the numerator we have the sum of the wages of all Italian provinces multiplied by their population, and at the denominator we have the entire Italian population of working age. As for the Status quo, this formula on Excel takes this form for the English version of Excel: "=(((SUM(U2:U63))*C8) + ((SUM(K2:K42))*M44)) / (SUM(SUM(U2:U63), SUM(K2:K42)))" (for Italian excel version we use "=(((SOMMA(U2:U63))*C8)+((SOMMA(K2:K42))*M44))/(SOMMA(SOMMA(U2:U63);SOMMA(K2:K42)))"). For the counterfactuals, changes "M" with "N" and "O".

between the values I found and the values in table 7, which could be due to statistical noise in the data.

For what concern the second panel on employment rate, instead, we have an discrepancy on the calculation of the counterfactual employment rate. In the paper by Boeri et al (2021, p.38), it is stated in the formula that the elasticity of the non-employment rate on value added is used to calculate the initial interest rate, i.e. 0.252. However, it appears in the code that it uses the elasticity of the employment rate, which is 0.1185. Below is the output of the Stata code that confirms the use of the last-mentioned value:

Figure C2.2 Elasticity of employment rate with respect to value added

lempl	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
lgva	.1184759	.0127816	9.27	0.000	.093403	.1435488
year						
2002	-.0129425	.0102829	-1.26	0.208	-.0331139	.0072289
2003	-.0203026	.0102901	-1.97	0.049	-.0404881	-.0001172
2004	-.033546	.0103092	-3.25	0.001	-.0537689	-.0133232
2005	-.0065259	.0103222	-0.63	0.527	-.0267742	.0137225
2006	.0206896	.0103666	2.00	0.046	.0003541	.041025
2007	.0400989	.0104093	3.85	0.000	.0196796	.0605181
2008	.0622575	.0104309	5.97	0.000	.0417959	.0827191
2009	.0696894	.0103525	6.73	0.000	.0493816	.0899971
2010	.084506	.0104491	8.09	0.000	.0640087	.1050033
2011	.0882851	.0105309	8.38	0.000	.0676273	.1089429
2012	.0897974	.0105682	8.50	0.000	.0690665	.1105284
2013	.0929905	.0106328	8.75	0.000	.0721329	.1138481
2014	.0934371	.0107254	8.71	0.000	.0723979	.1144763
2015	.0895707	.0108341	8.27	0.000	.0683181	.1108233
_cons	2.904462	.1371938	21.17	0.000	2.635338	3.173586

```
local slope_empl=_b[lgva]

31. replace empl_opt2=(1-`slope_empl'*((`max_gva'-`gva_var')/`max_gva'))*`max_empl' if top==0
```

In the Excel file, we experiment the calculation with both values and values are closer to the table 7 using the value elasticity of the employment rate with respect to the value added, than using the non-employment rate. In fact, if we used the value of the elasticity of non-employment (0.252), the counterfactual employment rate would be 68.76% (Column L of the second spreadsheet). Whereas if one used the elasticity of employment (0.118), one would have a counterfactual employment rate of 70.07%, which is much closer to the value in Table 7.

The calculation methods are similar to the Excel sheet described above, so tedious repetition is avoided. In the third sheet there are the calculations to find the aggregate labour income per capita, where the formula in subsection 5.3 is simply applied. Table C1 shows the results of my counterfactual calculations. Some small differences can be seen, but the substance remains the same. The conclusions are not altered and therefore these discrepancies are not a problem for our research. According to my results, aggregate employment and income costs are 8.61% and 5.70%, respectively.

Table C2. Counterfactual exercise for Italy independently calculated.

	South		North		Italy		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Level	Change	Level	Change	Level	Change	%Change
Average hourly wage *in euros*			Median				
Status quo	8.36		8.65		8.53		
Counterfactual I	7.84	-0.52	8.65	0.00	8.30	-0.23	-2.67
Counterfactual II	7.56	-0.80	8.65	0.00	8.17	-0.36	-4.18
Employment rate: *in % corrected for informal work*							
Status quo	57.32		71.24		64.99		
Counterfactual I	70.07	12.82	71.24	0.00	70.58	5.59	8.61
Counterfactual II	71.24	13.92	71.24	0.00	71.24	6.25	9.62
Aggregate labour income per capita: *in euros per month*							
Status quo	766.82		986.30		886.75		
Counterfactual I	879.45	112.63	986.30	0.00	937.29	50.55	5.70
Counterfactual II	861.49	94.67	986.30	0.00	931.34	44.60	5.03